Chironix: Turning to Google Cloud to help Aboriginal Australians connect with their ancestors

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About Chironix

Headquartered in Western Australia, Chironix is a software development company that focuses on robotic autonomy and AI. The business operates across three markets: the use of wearable devices—complementing the Internet of Things; building robot operating systems for industry verticals such as defense, resources, and health; and AI and machine learning across sectors such as genomics.

Industries: Technology

Google Cloud results

- Delivers the flexibility and scalability to enable researchers to complete analyses and provide detailed genetic summaries of project participants
- Reduces time to complete analysis and outputs from several months to weeks
- Provides a highly secure environment for sensitive genetic data

Location: Australia

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Google Cloud (https://cloud.google.com/)

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Compute Engine (https://cloud.google.com/compute/)

Cloud Life Sciences (https://cloud.google.com/life-sciences/)

For many Aboriginal Australians, learning about their ancestors' experiences and reconnecting with their ancestral homelands helps establish identity and provides context to their lives. Chironix (https://chironix.com/) is working with the University of Adelaide, the South Australian Museum, and Google Cloud on a project—called the Aboriginal Heritage Project—to enable Aboriginal Australians to learn more about their ancestors and cultural heritage.

The project is one of many innovative, research-based activities being undertaken by Chironix. The software development company, which focuses primarily on human-robot interactions and the use of technology to improve human potential, counts cloud-based, machine-learning-powered genomics research among its three key lines of business. The other two are wearable devices, with Chironix being Australia's only Glass for Enterprise partner, and building robot operating systems with applications across verticals such as defense, resources, and health.

Chironix's track record of innovation and achievement shows the business is well placed to take on the Aboriginal Heritage Project. The organization's ability to utilize and manage large datasets to achieve a business outcome has been the basis of its growth, and is set to continue to bring it into opportunities of significance moving forward.

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>
> *—Daniel Milford, Managing Director, Chironix*

Genomics an exciting field of development

“Genomics is a particularly dynamic and interesting field of medical development,” says Daniel Milford, Managing Director of Chironix. “Google Cloud, the optimization of high performance computing, and new processors are effectively transforming the field.” Chironix proved the case for using the cloud to work faster and at a lower cost with datasets associated with viruses; oats, barley, and rye; and the human genome.

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Managing datasets of 1.3 PB

To analyze individuals’ genome structures, Chironix turned to a Genome Analysis Toolkit (GATK) pipeline from Google Cloud. This pipeline uses the Cloud Life Sciences API (/life-sciences) and GATK best practices provided by the Broad Institute, a biomedical and genomic center based in the United States. Genome structure data is captured in files of about 350 GB.

Chironix uses preemptible virtual machine instances through Compute Engine (/compute) to provide compute resources and cold storage to manage the data generated from its genomics analysis, while machine learning models help reduce the time needed for discovery of the datasets to months, rather than years. This architecture can scale to manage datasets of 1.3 PB and beyond.

Reconstructing Aboriginal Australian history

The Aboriginal Heritage Project has particular significance for Aboriginal Australians affected by Stolen Generations policies—the removal by governments, churches, and welfare agencies of Aboriginal and Torres Strait Islander children from parents and communities. The Australian Institute of Aboriginal and Torres Strait Islander Studies (https://aiatsis.gov.au/) notes that governments as well as churches, missions, and non-government agencies lost many Stolen Generations records (https://aiatsis.gov.au/research/finding-your-family/before-you-start/stolen-generations) deliberately or through fire, flood, or poor practices.

Chironix is collaborating with a team of researchers and Aboriginal consultants at the Australian Centre for Ancient DNA (ACAD) and South Australian Museum to build a comprehensive genetic map of Aboriginal Australia. The map is based on a unique collection of Aboriginal Australian hair samples and genealogical information obtained with permission from more than 5,000 Aboriginal and Torres Strait Island people during anthropological expeditions undertaken between 1925 and 1963. Because these expeditions took place while European colonial disruption to Aboriginal Australian social organization was still taking effect, the collection provides a valuable link to the pre-colonial past.

“Importantly, this could provide some Aboriginal Australians impacted by colonial era policies with a first glimpse into the location of their ancestral homelands,” says Dr Yassine Souilmi, a computational geneticist working on the project.

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> provide detailed genetic summaries of each of the participants in the project, while obtaining a bigger picture view of Aboriginal Australian genetic history."
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> *—Dr Yassine Souilmi, Computational Geneticist, Aboriginal Heritage Project*

To build the genetic map, Aboriginal consultants contacted donors of the original hair samples or their living descendants to seek informed consent for use of the sample and related genealogical data. "Participating families identify having direct control over the consent process, and the opportunity to gain a new and valuable understanding of their past, as key reasons for getting involved," says Dr Ray Tobler, a geneticist working on the project.

Implementing genomics pipelines in Australia

“Google Cloud allows us to implement our genomics pipelines in a highly secure environment located in Australia. The platform is flexible and scalable, allowing us to complete our analyses and provide detailed genetic summaries of each of the participants in the project, while obtaining a bigger picture view of Aboriginal Australian genetic history,” says Dr Souilmi. “We can do this in a matter of weeks rather than several months.”

The final stage of the project entails enabling any Aboriginal Australian interested in learning more about their ancestry to compare their own raw DNA data against the genetic reference map developed from the hair samples.

“Using open source genome analytical toolkits and cloud computation through Google Cloud, sequencing the catalog of samples is possible and affordable."

—Daniel Milford, Managing Director, Chironix

"When the program is complete, a modern Aboriginal Australian can test a sample of their own DNA and reference it against this extensive genetic map," says Milford. "Aboriginal Australians will be able to identify and reconnect with their ancestors' country and their cultural heritage."

"When Norman Tindale collected hair samples from Aboriginal Australians as part of his anthropological expeditions in the early 20th century, he had no way of knowing that almost a century later these samples would be the key to allowing these Australians to reconnect with their heritage and homelands," adds Milford. "The scientific community did not fully understand DNA when Tindale was collecting his samples, and that they were collected at all was fortunate."

"Now, using open source genome analytical toolkits and cloud computation through Google Cloud, sequencing the catalog of samples is possible and affordable."

This project effectively gives back history, and knowledge of ancestry, to Aboriginal Australians affected by the policies of the Stolen Generations.

Chironix is able to take the building blocks of past research to work with Google Cloud to make the database possible—turning ancient DNA, 20th-century samples, and cutting-edge research and technology into a valuable resource that will empower Aboriginal Australians to connect with the past.

Daniel Milford, Managing Director, Chironix

Learn More

Reach out to our team to see how Google Cloud can help your business.

(https://cloud.google.com)